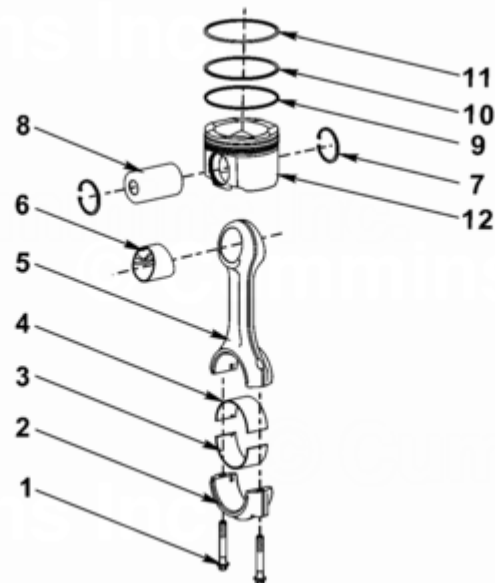


Trainings ▾

Exploded View



Piston and Connecting Rod Assembly Exploded View

1. Connecting rod capscrew - quantity 12
2. Connecting rod cap - quantity 6
3. Lower connecting rod bearing shell - quantity 6
4. Upper connecting rod bearing shell - quantity 6
5. Connecting rod - quantity 6
6. Connecting rod bushing - quantity 6
7. Retaining ring - quantity 12
8. Piston pin - quantity 6
9. Oil piston ring - quantity 6
10. Compression piston ring - quantity 6
11. Compression piston ring - quantity 6
12. Piston - quantity 6

Select Service Tools

Recommended Cummins® Service Tools

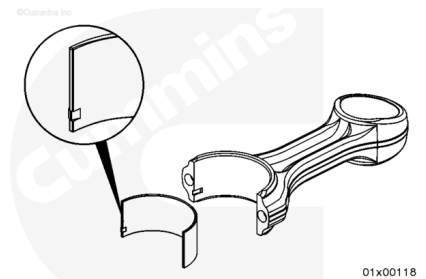
- Scotch-Brite™ 7448, Part Number 3823258, or equivalent
- Guide pin, Part Number 5298634
- Lubriplate™, Part Number 3163086
- Lubriplate™ 105, Part Number 3163087, or equivalent
- Piston ring compression band and pliers, Part Number 3164604

Additional Service Items

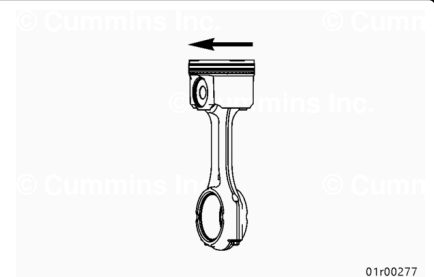
- Rubber hammer
- Dykem™
- Internal snap ring plier
- Blunt tool.

General Information

The connecting rod is a fracture-split connection between the rod cap and the rod. Connecting rods with a fracture-split surface **must** be treated with caution. The two pieces of the connecting rod **must not** be rubbed together. The mating surfaces will be damaged. Use care **not** to drop either piece of the connecting rod. Fracture-split connecting rods **must only** be handled if the two pieces of the connecting rod are tightened to the correct specification, or completely separated.

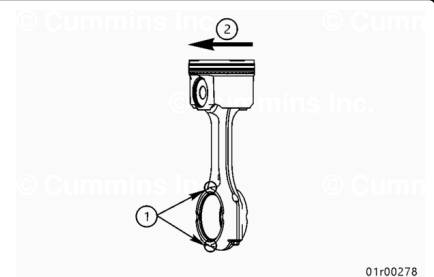


Each piston has an arrow on the top that points toward the front of the engine.



Piston and connecting rod orientation.

1. Pad orientation features on connecting rod points to piston front.
2. Arrow points to front of engine.



Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

WARNING

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

WARNING

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [122°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

WARNING

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

WARNING

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

CAUTION

Do not rotate the crankshaft with loose or missing connecting rod caps. Damage to the piston cooling nozzles can occur.

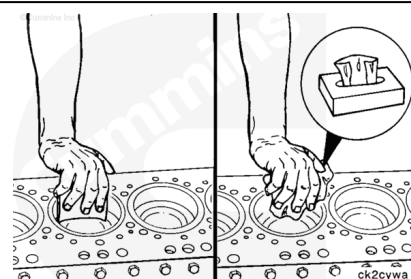
- Disconnect the batteries. See equipment manufacturer service information.
- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/1016/1016-007-037.html)

- Drain the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/1016/1016-008-018.html)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7.
(/qs3/pubsys2/xml/en/procedures/1016/1016-007-025.html)
- Remove the block stiffener plate. Refer to Procedure 001-089 in Section 1.
(/qs3/pubsys2/xml/en/procedures/1016/1016-001-089.html)
- Remove the piston cooling nozzles. Refer to Procedure 001-046 in Section 1.
(/qs3/pubsys2/xml/en/procedures/1016/1016-001-046.html)
- Remove the cylinder head. Refer to Procedure 002-004 in Section 2.
(/qs3/pubsys2/xml/en/procedures/1016/1016-002-004.html)

Remove

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

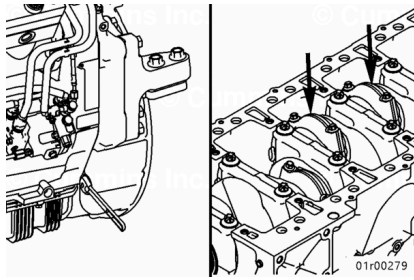
Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ CAUTION ⚠

Do not use emery cloth or sandpaper to remove carbon from the cylinder liners. Aluminum oxide or silicon particles from emery cloth or sandpaper can cause serious engine damage. Do not use any abrasives in the ring travel area. The cylinder liner can be damaged.

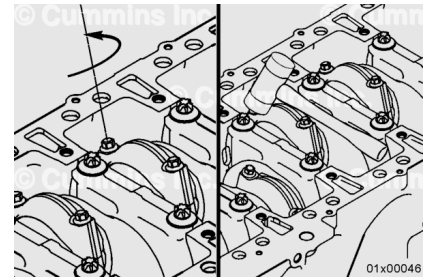
Use a fine, fibrous abrasive pad such as Scotch-Brite™ 7448, Part Number 3823258, or equivalent, and solvent to remove the carbon.

Rotate the crankshaft so that two of the connecting rods are at bottom dead center. Refer to Procedure 000-017 in Section 0.
(/qs3/pubsys2/xml/en/procedures/1016/1016-000-017.html)



⚠ CAUTION ⚠

Use care not to damage the fracture-split mating face(s) on the connecting rod and connecting rod cap while the connecting rod caps are removed. If the fracture-split surface is damaged, the connecting rod and connecting rod cap must be replaced to reduce the possibility of engine damage. Incorrect assembly can also damage the connecting rod and connecting rod cap.



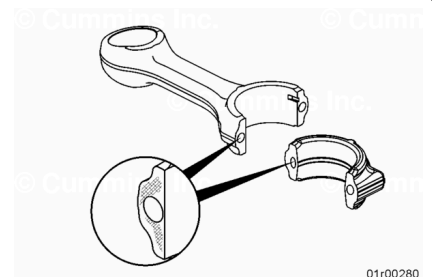
Do **not** remove the capscrews at this time.

Loosen the connecting rod capscrews.

Hit the connecting rod capscrews with a rubber hammer to loosen the connecting rod caps from the dowels.

⚠ CAUTION ⚠

To prevent damage to the fractured-split connecting rod and connecting rod cap, do not set the connecting rod or connecting rod cap on the fracture-split mating face(s). Polishing and damage to the mating surface(s) can result.



⚠ CAUTION ⚠

Minimize the number of joint interactions between the connecting rod and rod cap. If the fracture-split mating surface becomes damaged or worn, the rod cap will not fit as designed to the connecting rod and engine damage can occur.

Remove the capscrews and the rod cap. Note the corresponding rod and rod cap as well as the orientation of the rod cap.

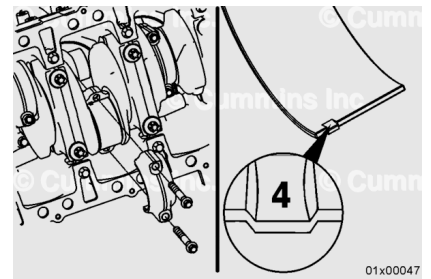
Use care **not** to damage the fracture-split connection.

Do **not** set the rod cap down on the fracture-split face.

Both the connecting rod cap and the connecting rod are serialized on one face to aid in proper orientation.

Connecting rod shell bearing upper or lower location is denoted by UPPER or UPR (for connecting rod) and LOWER or LWR (for connecting rod cap) stamped into the back of the bearing shells.

Remove the bearing shell from the rod cap and mark the cylinder number on the flat surface of the bearing tangs.

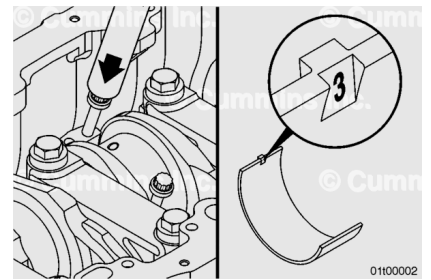


Use a connecting rod guide pin, Part Number 5298634, to push the rod up far enough to allow the upper bearing shell to be removed.

Do **not** push against the fracture-split face of the rod.

Use caution **not** to scratch or damage the journal with the rod, capscrew, or piston pusher.

Remove the bearing shell and mark the cylinder number on the flat surface of the bearing tangs.



⚠ CAUTION ⚠

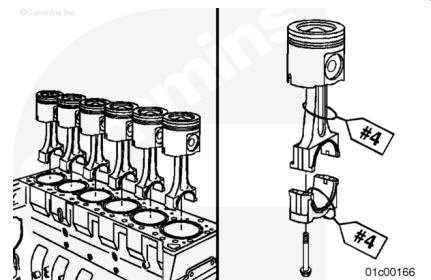
Avoid contact between rod and liner upon removal of connecting rod assembly to avoid damaging the liner surface.

Remove the piston and connecting rod assembly.

The piston and connecting rod assemblies **must** be installed in the same cylinder from which the piston and connecting rod assemblies are removed.

Use a tag to mark the cylinder number from which each piston and connecting rod assembly is removed.

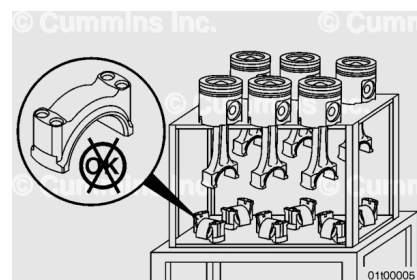
Pistons that are reused **must** be installed in the same orientation as when removed from the engine. Mark the front side of the piston and piston pin with Dykem™.



Place the connecting rod and piston assemblies in a container to protect them from damage.

The two mating faces of the fracture-split connecting rod can **not** be rubbed together. The mating surfaces will be damaged.

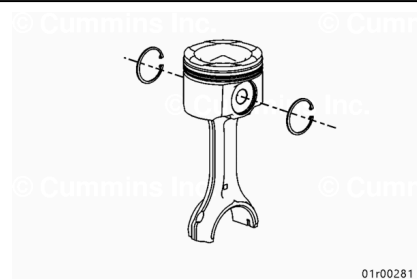
Use care **not** to drop either piece of the connecting rod. Fracture-split connecting rods **must only** be handled if the two pieces of the connecting rod are tightened to the correct specification or completely separated.



Disassemble

Use internal snap ring pliers to remove the snap rings from both sides of the piston.

If the piston is being replaced, the piston pin **must** also be replaced.

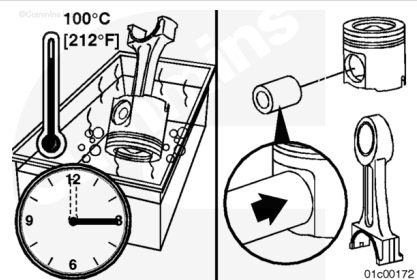


⚠ CAUTION ⚠

Do not use a hammer to remove the piston pins. The piston can distort, causing the piston to seize in the liner.

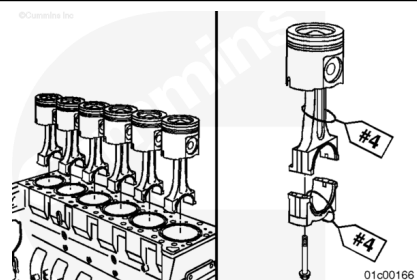
If the piston pin can **not** be easily removed by hand, place the piston and the connecting rod assembly in a container of water. Heat the piston in boiling water for 15 minutes.

Use a blunt tool to push the piston pin from the piston and connecting rod assembly.



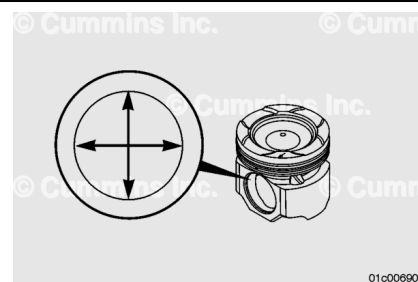
The piston and connecting rod assemblies **must** be installed in the same cylinder from which the piston and connecting rod assemblies are removed to provide for proper fit of worn mating surfaces if parts are reused.

Use a tag to mark the cylinder number from which each piston and connecting rod assembly is removed.

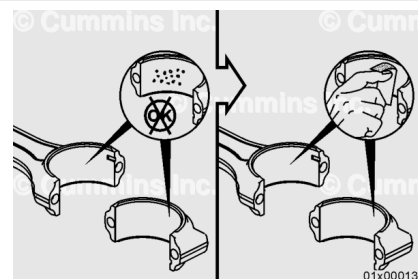


Clean and Inspect for Reuse

Clean and inspect the pistons and piston pins. Refer to Procedure 001-043 in Section 1.
(/qs3/pubsys2/xml/en/procedures/1016/1016-001-043.html)



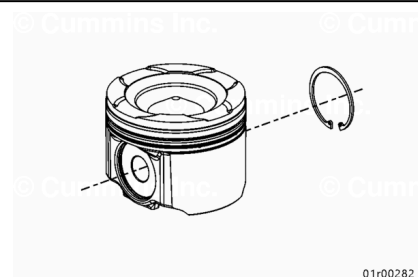
Clean and inspect the connecting rods. Refer to Procedure 001-014 in Section 1.
(/qs3/pubsys2/xml/en/procedures/1016/1016-001-014.html)



Assemble

⚠ CAUTION ⚠

The retainer snap ring must be seated completely in the piston pin groove to prevent engine damage.



Do **not** reuse piston pin snap rings.

Install a new snap ring into one piston pin bore of each piston with the gap of the snap ring at the top or bottom of the piston pin bores.

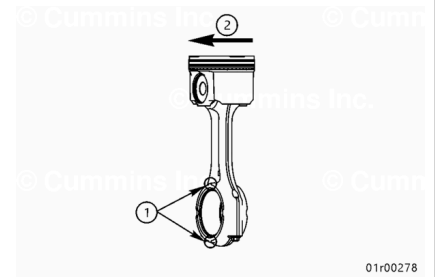
If the pistons are being reused, the piston and pin **must** be matched as when removed.

The pistons have a directional arrow etched into the top of the piston crown. Fracture-split connecting rods have pad orientation features toward piston front.

Piston and connecting rod orientation.

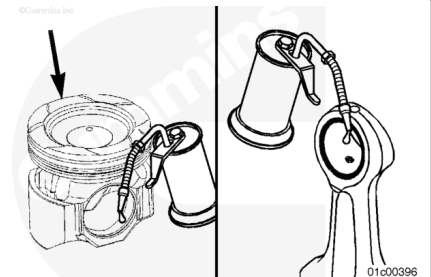
1. Pad orientation features on connecting rod points to piston front.

2. Arrow points to front of engine.



⚠ CAUTION ⚠

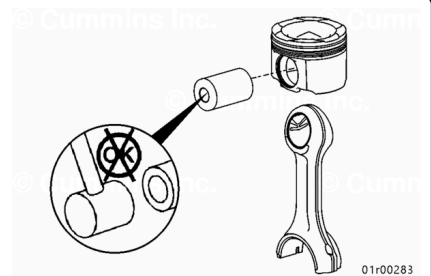
Generously coat both piston pin bores with a heavy weight gear oil, such as 85W-140 EP, when assembling the pistons. Also lubricate the connecting rod bushing. If gear oil is not available, Lubriplate™, Part Number 3163086, is permissible. Inadequate lubrication of the pin joint can result in extensive engine damage.



Use clean 85W-140 oil to lubricate piston pin, piston pin bore, and connecting rod bushing. The lubrication shall cover a minimum of 270 degrees of circumference and include the loaded region of the pin bores.

⚠ CAUTION ⚠

Do not use a hammer to install the piston pin. The piston can distort, causing the piston to seize in the liner.

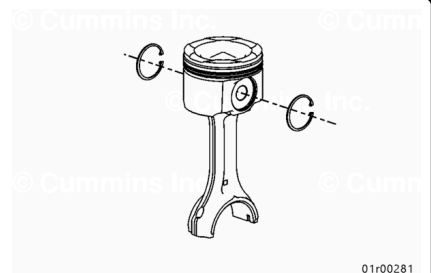


For pistons that are being reused, the piston, piston pin, and connecting rod **must** be installed in the same orientation as when removed from the engine.

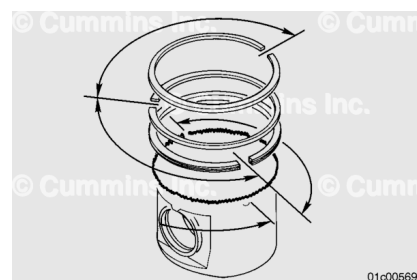
Pistons do **not** require heating to install the pin. However, the pistons do need to be at room temperature or above.

Install the pin through the piston and connecting rod.

Install a new snap ring into the piston pin bore with the snap ring gap at the top or bottom of the piston pin bore.



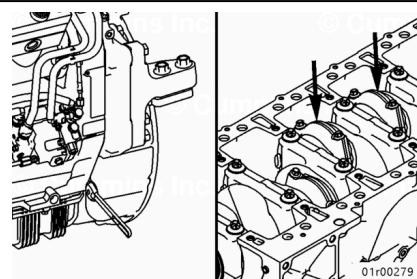
Install the piston rings. Refer to Procedure 001-047 in Section 1. (</qs3/pubsys2/xml/en/procedures/1016/1016-001-047.html>)



Install

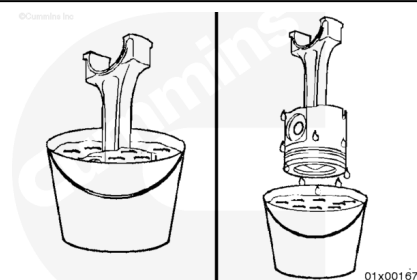
Rotate the crankshaft so the connecting rod journal of the connecting rod being installed is at bottom dead center. Refer to Procedure 000-017 in Section 0.

(</qs3/pubsys2/xml/en/procedures/1016/1016-000-017.html>)



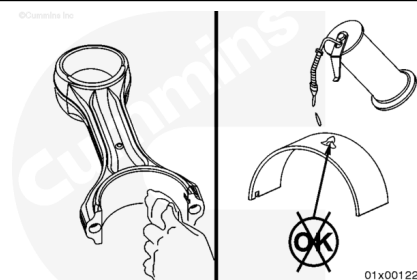
Immerse the piston in a container of clean engine oil.

Remove the piston from the container. Let the excess oil drain from the piston.



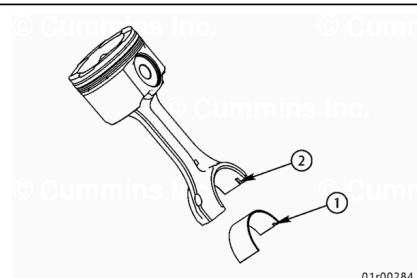
Use a clean, lint-free cloth to clean the connecting rods and bearing shells.

Do **not** lubricate the back side of the bearing shells. The operating clearance of the bearing will be reduced and the bearing can be damaged during engine operation.



If new bearings are **not** used, the used bearings **must** be installed on the same connecting rod from which the bearings are removed.

Install the upper bearing shell into the connecting rod.

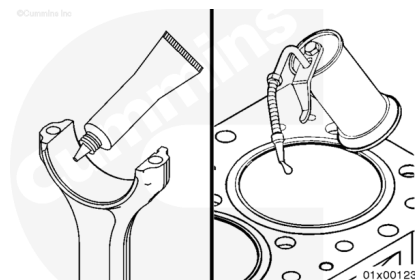


The tang of the bearing shell (1) **must** be in the slot of the connecting rod (2). The end of the bearing shell **must** be even with the cap mounting surface.

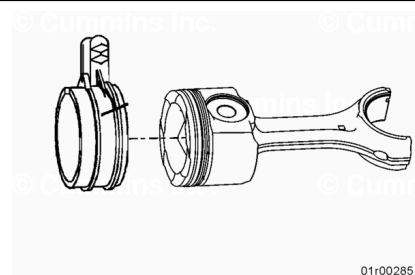
Use Lubriplate™ 105, Part Number 3163087, or equivalent, to coat the entire inside circumference of the bearing shell.

Inspect the cylinder liners for reuse. Refer to Procedure 001-028 in Section 1.
(/qs3/pubsys2/xml/en/procedures/1016/1016-001-028.html)

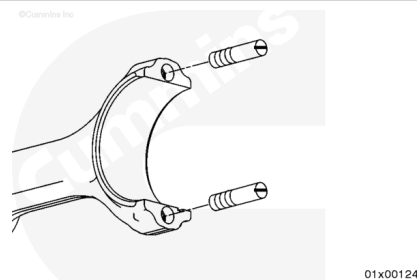
Apply a film of clean engine oil to the cylinder liner.



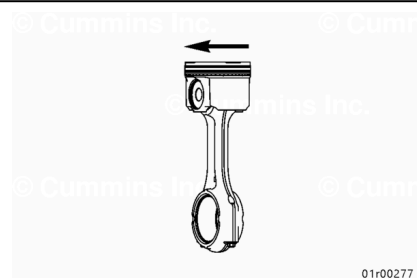
Use tool piston ring compression band and pliers, Part Number 3164604, to compress the rings. The cylinder block and all parts **must** be clean before assembly.



Install two guide pins, Part Number 5298634, hand-tight in the connecting rod to minimize the risk of damaging the fracture-split surface.



Verify the arrow on the top of piston points to front of engine.



⚠ WARNING ⚠

To reduce the possibility of personal injury or equipment damage, care must be taken during the installation of the connecting rod and piston assembly into the cylinder liner.

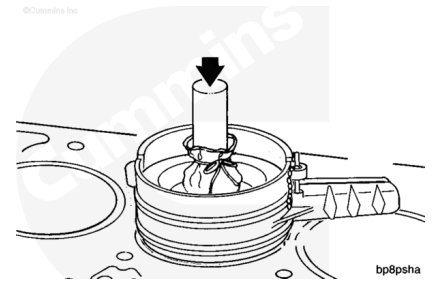
The assembly can fall into the cylinder liner abruptly, once the piston rings clear the piston ring compressor as it enters the cylinder liner.

⚠ CAUTION ⚠

Do not use a metal drift to push the piston into the cylinder liner. The piston rings or cylinder liner can be damaged.

⚠ CAUTION ⚠

The connecting rod and piston assembly will fall into the cylinder once the rings have cleared the bottom of the ring compressor. The fracture-split surface and crankshaft journal bearing surface can be damaged if allowed to contact the crankshaft.

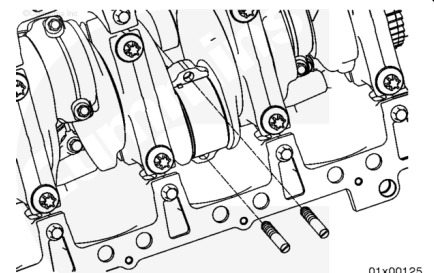


Hold the ring compressor against the cylinder liner. Push the piston through the ring compressor and into the cylinder liner. Push the piston until the top ring is completely in the cylinder liner.

If the piston does **not** move freely, remove the piston and inspect for broken or damaged rings.

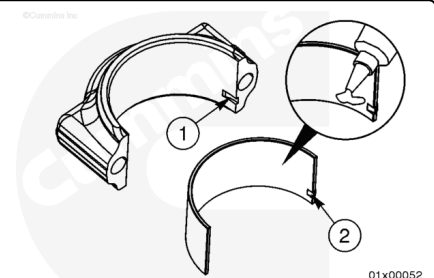
Align the connecting rod with the crankshaft while pushing the piston and connecting rod assembly into place.

Once the rod is seated in place, remove the guide pins.

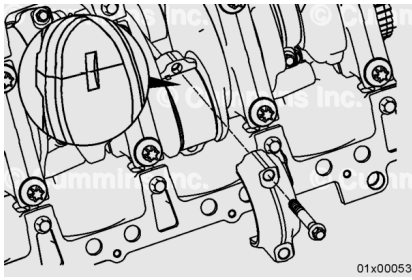


Install the lower bearing shell into the connecting rod cap with the tang (2) of the bearing in the slot (1) of the cap.

Use clean Lubriplate™ 105 multi-purpose lubricant, or equivalent, to lubricate the bearing shell to crankshaft journal mating surface and the connecting rod capscrews.



Install the cap onto the connecting rod.



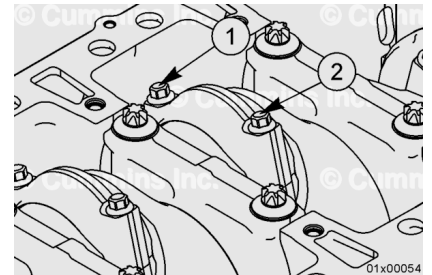
Lubricate the threads and underside of the heads of the connecting rod capscrews with clean engine oil.

Thread the capscrews into the connecting rod and hand tighten.

Torque Value:

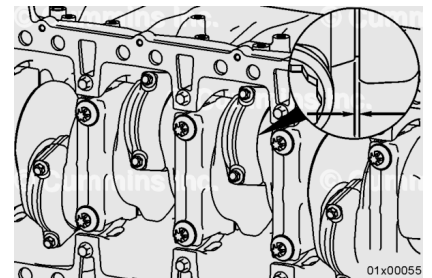
1. 91 n•m [67 ft-lb]

2. Angle 60 degrees.



Measure the connecting rod side clearance.

Connecting Rod Side Clearance		
mm		in
0.10	MIN	0.004
0.35	MAX	0.014



The connecting rod **must** move freely from side to side on the crankshaft journal. If the connecting rod does **not** move freely, remove the connecting rod cap and verify the bearing shells are the correct size. Check for dirt or damage on the crankshaft and the bearing shells.

Repeat the above steps to install the remaining bearing shells and connecting rod caps.

Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the piston cooling nozzles. Refer to Procedure 001-046 in Section 1.
(/qs3/pubsys2/xml/en/procedures/1016/1016-001-046.html)
- Install the block stiffener plate. Refer to Procedure 001-089 in Section 1.
(/qs3/pubsys2/xml/en/procedures/1016/1016-001-089.html)
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7.
(/qs3/pubsys2/xml/en/procedures/1016/1016-007-025.html)
- Install the cylinder head. Refer to Procedure 002-004 in Section 2.
(/qs3/pubsys2/xml/en/procedures/1016/1016-002-004.html)
- Prime and fill the lubricating oil system. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/1016/1016-007-037.html)
- Fill the cooling system with coolant. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/1016/1016-008-018.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine to normal operating temperature and check for leaks and proper operation.

All engines **must** be run-in after a rebuild or a repair involving the replacement of one or more piston ring sets, cylinder liners, or pistons. See Section 14 - Engine Testing for the appropriate run-in procedure.

Use the following procedure for a general run-in test overview. Refer to Procedure 014-999 in Section F. (/qs3/pubsys2/xml/en/procedures/1016/1016-014-999.html)

If a malfunction results in coolant, oil, excessive fuel, or excessive black smoke entering the exhaust system, the aftertreatment system **must** be inspected.